English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Stream 2 Written examination, 30 June 2022 – *Your grade is 2 + the number of points. Time allowed: 3 hours. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$*, with random variables* X, Y*. If needed, use*

$$\int e^{-x}\,dx = -e^{-x}, \quad \int xe^{-x}\,dx = -e^{-x}(x+1), \quad \int x^2e^{-x}\,dx = -e^{-x}(x^2+2x+2).$$

Problem 1. A magic square is a 3×3 table with equal sums along each row, column and the two main diagonals. A fifth-grader must fill the following square with the numbers 1 through 9 exactly once:

2	7	
	5	
		8

Unwilling to add numbers, the pupil writes a program to fill the blanks randomly until it finds the correct configuration. **One simulation independently chooses five uniform numbers from** $\{1, \ldots, 9\}$ **and fills the blank cells from top to bottom and left to right.**

1. What is the expected number of correct entries in each simulation?
2. What is the expected number of simulations until the correct arrangement?
3. Suggest n such that the probability of at least one success in n simulations is 99%.

Answer the same questions if **a simulation is a random ordering of the missing numbers** $(1, 3, 4, 6, 9)$.

Problem 2. Each face of a blank die is labelled with a random integer from 1 to 6. Roll it n times. Let X sum the first $n-1$ rolls and Y the last $n-1$. Find $\mathrm{Cor}(X, Y)$. How would the answer change if labels were chosen between 199 and 999?

Problem 3. A charity recruits volunteers to collect donations. Each talks to 100 people daily; on average one in 25 donates a fixed 10 BGN. What is the smallest number of volunteers you would recruit so the probability of raising at least one million BGN in 100 days is at least 99%?[1] With that many volunteers, what is the probability of raising the money in 99 days?

Problem 4. A mechanism has two components with lifetimes X, Y hours and joint density $f_{X,Y}(x, y) = cxe^{-(x+y)/2}$ for $x, y > 0$, and 0 otherwise. Find:

1. c;
2. the probability that both work for more than one hour;
3. $\mathbb{E}[Y/X]$;
4. $\mathbb{E}[X \mid Y = 1]$.

Are X, Y independent?

[1] Chebyshev's inequality is too crude to be suitable here.